Example 8.2

If $\vec{a}$ and $\vec{b}$ are vectors represented by two adjacent sides of a regular hexagon, then find the vectors represented by other sides.

Solution

Let A,B,C,D,E,F be the vertices of a regular hexagon.

Let $\vec{a} = \overrightarrow{AB}$ and $\vec{b} = \overrightarrow{BC}$

We use the following facts about regular hexagon.

(i) The lines AB, CF and ED are parallel and the lines BC, AD and EF are parallel.

(ii) The length of CF is twice the length of AB and the length of AD is twice the length of BC.

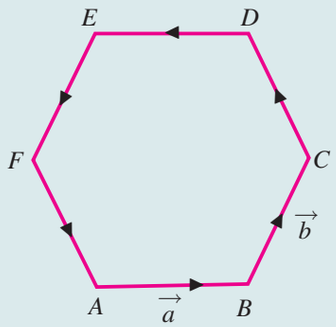


Fig. 8.25

Since the lines AB and DE are parallel, equal in length and opposite in direction we have

$$\overrightarrow{DE} = -\vec{a}\ .$$

Since the lines AB and CF are parallel and opposite in direction we have

$$\overrightarrow{CF} = -2\vec{a}.$$

Similarly $\overrightarrow{EF} = -\vec{b}$ and $\overrightarrow{AD} = 2\vec{b}$.

Since $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ we have

$$\overrightarrow{AC} = \vec{a} + \vec{b}.$$

Since $\overrightarrow{AC} + \overrightarrow{CD} = \overrightarrow{AD}$ we have

$$\vec{a} + \vec{b} + \overrightarrow{CD} = 2\vec{b}.$$

Thus $$\overrightarrow{CD} = 2\vec{b} - (\vec{a} + \vec{b}) = \vec{b} - \vec{a}.$$

As $\overrightarrow{FA} = -\overrightarrow{CD}$, we have

$$\overrightarrow{FA} = \vec{a} - \vec{b}.$$

Hence, for given sides $\overrightarrow{AB} = \vec{a}$ and $\overrightarrow{BC} = \vec{b}$, we have obtained all other sides of the hexagon as $\overrightarrow{CD} = \vec{b} - \vec{a}, \overrightarrow{DE} = -\vec{a}, \overrightarrow{EF} = -\vec{b}$, and $\overrightarrow{FA} = \vec{a} - \vec{b}$.

8.5 Position vectors

Definition 8.14

Let O be the origin and P be any point (in the plane or space). Then the vector $\overrightarrow{OP}$ is called the **position vector** of the point P with respect to the origin O (point of reference).

The relation between the vectors and position vectors are given in the following result.

Result 8.6

Let O be the origin, A and B be two points. Then $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$ where, $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are position vectors of A and B respectively.

Proof

We know that, $\overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$. Thus $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$.

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